Equation sheet: governing relations of the $CaCO_3$-barrier screen

P0-W3.1. Symbolic only; every symbol is in the table overleaf and in `theory/symbols.yaml`; units in brackets.

	R-slurry: aqueous, alkaline, room temperature; casting and drying on native Al_2O_3/AlOOH	R-cell: non-aqueous, fluorine-free electrolyte, collector at anodic potential vs Li^+/Li
Links that apply	C1 adhesion (1), C2 sequestration (2), supersaturation (3), C3 nucleation and Young (4, 5), template action (6), C4 morphology (7), C5 aqueous Pourbaix (8a)	C5 Li-open stability window (8b), C4 insulating-film bound (9). Not defined here: (1)–(7), since there is no hydrated Ca^{2+} reference and no aqueous Ω

1 Adhesion in water (C1, R-slurry)

The polymer never meets a dry oxide: every contact first displaces interfacial water, so the decisive work separates the contact into two water-wetted *surfaces.*

$$W_{\mathrm{adh,wet}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}] \tag{1}$$
$$= W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}$$
$$W_{\mathrm{adh,dry}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]$$
$$W_{\mathrm{adh,wet}} = \Delta F_{\mathrm{PMF}}/A_{\mathrm{c}} \quad [\mathrm{J\,m^{-2}}] \tag{2}$$

Anchored against water iff $W_{\mathrm{adh,wet}} > 0$; $W_{\mathrm{ow}}, W_{\mathrm{pw}}$ are the dry works of the oxide and polymer with water.

2 Sequestration (C2, R-slurry)

The template raises local Ca^{2+} only if it holds Ca^{2+} more strongly than water does; the cycle starts from hydrated Ca^{2+} at the regime activity.

Cycle: Ca^{2+}(aq, bulk, $a_{Ca^{2+}}$) $\rightarrow$ Ca^{2+}(binding layer). Solution side: molal standard state at the regime ionic strength; layer side: site-volume standard state, $\Delta G_{\mathrm{site}} = -RT\ln\big[V_{\mathrm{site}}^{-1}\int_{\mathrm{site}} \mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}\,\mathrm{d}^3r\big]$, with w the per-ion PMF ($w \rightarrow 0$ in bulk). With one coordinating group that must be deprotonated:

$$\Delta G_{\mathrm{seq}} = \Delta G_{\mathrm{site}} + RT\ln\big(1 + 10^{\,\mathrm{p}K_{\mathrm{a}} - \mathrm{pH}}\big) \quad [\mathrm{J\,mol^{-1}}] \tag{3}$$

$$\mathcal{E} \equiv \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}} = \exp\Big(-\frac{\Delta G_{\mathrm{seq}}}{RT}\Big) \quad [1] \tag{4}$$

Valid for dilute Ca^{2+} with unsaturated sites ($\mathcal{E}c_{\mathrm{bulk}} \ll c_{\mathrm{max}}$) and equal activity coefficients in bulk and layer (so $\mathcal{E}$ is both a concentration and an activity ratio). Saturation (Langmuir closure):

$$c_{\mathrm{local}} = \frac{\mathcal{E}c_{\mathrm{bulk}}}{1 + \mathcal{E}c_{\mathrm{bulk}}/c_{\mathrm{max}}} \le \min(\mathcal{E}c_{\mathrm{bulk}},\ c_{\mathrm{max}}) \quad [\mathrm{mol\,m^{-3}}] \tag{5}$$
$$c_{\mathrm{max}} = \Gamma_{\mathrm{s}}/\delta_{\mathrm{s}} \quad [\mathrm{mol\,m^{-3}}]$$

3 Supersaturation (one definition)

Crystallisation is driven by the excess chemical potential of the dissolved ion pair over the solid of the registered framing (ACC or calcite).

$$\Omega = \frac{\mathrm{IAP}}{K_{\mathrm{sp}}} = \frac{a_{Ca^{2+}}\,a_{CO_3^{2-}}}{K_{\mathrm{sp}}}, \qquad \Delta\mu = k_{\mathrm{B}}T\ln\Omega \quad [\mathrm{J}] \tag{6}$$

per formula unit; nucleation requires $\Omega > 1$ ($\Delta\mu > 0$). Only Ca^{2+} is enriched; carbonate stays at its bulk activity:

$$\Omega_{\mathrm{local}} = \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}}\,\Omega_{\mathrm{bulk}} \xrightarrow{\text{dilute}} \mathcal{E}\,\Omega_{\mathrm{bulk}} \quad [1] \tag{7}$$

$$\Delta\mu_{\mathrm{local}} = \Delta\mu_{\mathrm{bulk}} - \Delta G_{\mathrm{seq}}/N_{\mathrm{A}} \quad [\mathrm{J}] \tag{8}$$

4 Classical nucleation (C3, R-slurry)

A cluster pays surface free energy before its volume gains $\Delta\mu$; the barrier is the top of that trade-off. On a substrate the cluster is a spherical cap with the same curvature.

$$\Delta G^*_{\mathrm{hom}} = \frac{16\pi\,\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\,\Delta\mu^2}, \quad r^* = \frac{2\gamma_{\mathrm{nl}}v_{\mathrm{m}}}{\Delta\mu}, \quad v_{\mathrm{m}} = \frac{V_{\mathrm{m}}}{N_{\mathrm{A}}} \quad [\mathrm{J};\ \mathrm{m}] \tag{9}$$

$$\Delta G^*_{\mathrm{het}} = f(\theta)\,\Delta G^*_{\mathrm{hom}}, \qquad f(\theta) = \tfrac{1}{4}(2+\cos\theta)(1-\cos\theta)^2 \tag{10}$$

$f(0) = 0$, $f(\pi/2) = \frac{1}{2}$, $f(\pi) = 1$; $f'(\theta) = \frac{3}{4}\sin^3\theta \ge 0$, so $0 \le \Delta G^*_{\mathrm{het}} \le \Delta G^*_{\mathrm{hom}}$ on $[0,\pi]$. $\Delta G^*_{\mathrm{hom}} \rightarrow \infty$ as $\Omega \rightarrow 1^+$, and $\Delta G^*_{\mathrm{hom}} \rightarrow 0$ as $\Omega \rightarrow \infty$.

5 Young, with one sign convention

θ is measured through the nucleus. Forming the contact removes substrate–liquid interface and creates substrate–nucleus interface over the same disc; $\cos\theta$ is that energy gain per unit nucleus–liquid energy.

$$\cos\theta = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}} \quad [1;\ \mathrm{J\,m^{-2}}] \tag{11}$$

$\theta = 0$: the nucleus wets the template completely (no barrier); $\theta = \pi$: no wetting (homogeneous barrier). $\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)}$ is the free energy per unit contact area to displace the template-bound water layer; $\Delta\gamma_{\mathrm{w}} > 0$ **is a cost** and raises θ, f and $\Delta G^*_{\mathrm{het}}$ ($\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = -1/\gamma_{\mathrm{nl}} < 0$). If γ_{sn} is supplied as a full in-water interfacial free energy, $\Delta\gamma_{\mathrm{w}} \equiv 0$ and is not added again. $\gamma_{\mathrm{eff}} \equiv \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ (deck form) is **not** used: Young and $\Delta G^*_{\mathrm{hom}}$ take the bare γ_{nl}. Reason: γ_{nl} multiplies the cap's free surface, which never touches the template, and $\Delta G^*_{\mathrm{het}} = f\,\Delta G^*_{\mathrm{hom}}$ follows only if the same γ_{nl} sits in Young and in $\Delta G^*_{\mathrm{hom}}$. With γ_{eff} in the denominator, $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = (\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)})/\gamma_{\mathrm{eff}}^2 > 0$ on the wetting side: a cost would lower the barrier.

6 Where the template acts

The template enters the barrier through two channels only: wetting (interfacial energies, via θ) and Ca^{2+} enrichment (via Ω).

$$\Delta G^*_{\mathrm{het}} = f\big(\theta[\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}^{(0)}, \Delta\gamma_{\mathrm{w}}]\big)\ \frac{16\pi\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\big(k_{\mathrm{B}}T\ln\Omega_{\mathrm{local}}[\Delta G_{\mathrm{seq}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}}]\big)^2} \tag{12}$$

No-double-counting condition: each template-dependent input enters exactly one channel ($\partial\theta/\partial\Delta G_{\mathrm{seq}} = 0$, $\partial\Omega/\partial\gamma_{\mathrm{s\cdot}} = 0$), and $\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}$ are evaluated without the template-bound Ca^{2+} excess already credited to Ω_{local}. Bulk inputs ($\gamma_{\mathrm{nl}}, V_{\mathrm{m}}, T$) may enter both. Proof: P0-W3.2.

7 Conformality (C4, R-slurry)

A barrier needs a continuous film; whether the mineral spreads or beads is a wetting balance, not a barrier height.

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}] \tag{13}$$

$S_{\mathrm{spread}} \ge 0$: no finite θ ($\theta \rightarrow 0$, $f \rightarrow 0$), layer-by-layer growth. $S_{\mathrm{spread}} < 0$: $\theta > 0$, island growth; a film forms only on coalescence, at cap height $h_{\mathrm{c}} = \tan(\theta/2)/(2\sqrt{N_{\mathrm{s}}})$ [m]. A low f does not imply coverage: $f(\pi/2) = \frac{1}{2}$ is still island growth.

8 Stability (C5): screens, necessary not sufficient

(a) R-slurry, aqueous Pourbaix, per non-O, non-H atom [$\mathrm{J\,mol^{-1}}$]:

$$\Psi_\phi = \tfrac{1}{n_{\mathrm{M}}}\big[G_\phi - n_{\mathrm{O}}\mu_{\mathrm{H_2O}} - RT\ln 10\,(2n_{\mathrm{O}} - n_{\mathrm{H}})\,\mathrm{pH} - (2n_{\mathrm{O}} - n_{\mathrm{H}} + Q)FE_{\mathrm{SHE}}\big] \tag{14}$$

$\Delta G_{\mathrm{pbx}} = \Psi_\phi - \Psi_{\mathrm{hull}} \ge 0$; zero means stable at (pH, E_{SHE}).

(b) R-cell, Li open, E_{Li} vs Li^+/Li [$\mathrm{J\,mol^{-1}}$; V]:

$$\mu_{\mathrm{Li}} = \mu^\circ_{\mathrm{Li}} - FE_{\mathrm{Li}}, \qquad \Phi_\phi = G_\phi - n_{\mathrm{Li}}\mu_{\mathrm{Li}} \tag{15}$$

Window $[E_{\mathrm{red}}, E_{\mathrm{ox}}]$: the E_{Li} over which ϕ lies on the Li-open hull.

9 Insulating film (C4 limitation, R-cell)

The mineral is an electronic insulator, so conformality cannot be bought with thickness; the film must still pass the collector current.

$$R_A = \rho_{\mathrm{e}}d \le R_{A,\mathrm{max}} \ \Rightarrow\ d \le R_{A,\mathrm{max}}/\rho_{\mathrm{e}} \quad [\Omega\,\mathrm{m^2};\ \mathrm{m}] \tag{16}$$

$$j = j_0\mathrm{e}^{-2\kappa d}, \quad \kappa = \frac{\sqrt{2m_{\mathrm{e}}\phi_{\mathrm{B}}}}{\hbar} \ \Rightarrow\ d \le \frac{\ln(j_0/j_{\mathrm{req}})}{2\kappa} \quad [\mathrm{m}] \tag{17}$$

Symbols

generated from `theory/symbols.yaml`; Reg.: S = R-slurry, C = R-cell; – = dimensionless

Symbol	Meaning	SI unit	Reg.
Exact SI constants (kept symbolic)			
k_B	Boltzmann constant	$\mathrm{J\,K^{-1}}$	S,C
N_A	Avogadro constant	$\mathrm{mol^{-1}}$	S,C
R	molar gas constant	$\mathrm{J\,K^{-1}\,mol^{-1}}$	S,C
F	Faraday constant	$\mathrm{C\,mol^{-1}}$	S,C
e	elementary charge	C	S,C
$\hbar$	reduced Planck constant	J s	C
m_e	electron mass	kg	C
State variables			
T	absolute temperature of the regime	K	S,C
pH	pH (molal scale)	–	S
E_SHE	collector potential vs SHE	V	S
E_Li	collector potential vs $\mathrm{Li^+/Li}$	V	C
Adhesion in water (C1)			
γ_ow	oxide-water interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_pw	polymer-water interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_op	oxide-polymer interfacial free energy in contact	$\mathrm{J\,m^{-2}}$	S
γ_o	oxide surface free energy against vacuum	$\mathrm{J\,m^{-2}}$	S
γ_p	polymer surface free energy against vacuum	$\mathrm{J\,m^{-2}}$	S
γ_w	water surface free energy against its vapour	$\mathrm{J\,m^{-2}}$	S
$W_\mathrm{adh,wet}$	work of adhesion in water (Dupre)	$\mathrm{J\,m^{-2}}$	S
$W_\mathrm{adh,dry}$	work of adhesion against vacuum (comparator)	$\mathrm{J\,m^{-2}}$	S
ΔF_PMF	PMF free energy, separated minus bound	J	S
A_c	fragment contact area	$\mathrm{m^2}$	S
Sequestration (C2)			
ΔG_seq	$\mathrm{Ca^{2+}}$ transfer free energy, bulk to binding layer	$\mathrm{J\,mol^{-1}}$	S
ΔG_site	transfer free energy to the deprotonated site	$\mathrm{J\,mol^{-1}}$	S
$\mathrm{p}K_\mathrm{a}$	pKa of the coordinating group	–	S
$\mathcal{E}$	$\mathrm{Ca^{2+}}$ enrichment factor (dilute limit)	–	S
c_bulk	bulk $\mathrm{Ca^{2+}}$ concentration	$\mathrm{mol\,m^{-3}}$	S
c_local	mean $\mathrm{Ca^{2+}}$ concentration in the layer	$\mathrm{mol\,m^{-3}}$	S
c_max	saturation concentration of the layer	$\mathrm{mol\,m^{-3}}$	S
Γ_s	areal binding-site density	$\mathrm{mol\,m^{-2}}$	S
δ_s	binding-layer thickness	m	S
V_site	volume per binding site	$\mathrm{m^3}$	S
Supersaturation			
$a_{\mathrm{Ca}2+}$	bulk $\mathrm{Ca^{2+}}$ activity	–	S
$a_{\mathrm{CO_3^{2-}}}$	bulk carbonate activity	–	S
K_sp	solubility product of the registered framing	–	S
Ω_bulk	bulk saturation ratio IAP/Ksp	–	S
Ω	saturation ratio where the nucleus forms	–	S
$\Delta\mu$	driving force per formula unit	J	S
Nucleation and Young (C3)			
γ_nl	nucleus-liquid interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_sl	template-liquid interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_sn	template-nucleus energy in water (= ref + correction)	$\mathrm{J\,m^{-2}}$	S
$\gamma_\mathrm{sn}^{(0)}$	template-nucleus energy, reference without water displacement	$\mathrm{J\,m^{-2}}$	S
$\Delta\gamma_\mathrm{w}$	water-displacement correction (cost > 0)	$\mathrm{J\,m^{-2}}$	S
γ_eff	deck effective nucleus-liquid energy (audit only)	$\mathrm{J\,m^{-2}}$	S
θ	contact angle, measured through the nucleus	rad	S
$f(\theta)$	heterogeneous shape factor	–	S
V_m	molar volume of the registered framing	$\mathrm{m^3\,mol^{-1}}$	S
r^*	critical radius	m	S
ΔG^*_hom	homogeneous nucleation barrier per nucleus	J	S
ΔG^*_het	heterogeneous barrier on the template	J	S
Conformality (C4)			
S_spread	spreading coefficient	$\mathrm{J\,m^{-2}}$	S
N_s	island density at coalescence	$\mathrm{m^{-2}}$	S
h_c	geometric coalescence thickness	m	S
h_cl	geometric closure thickness	m	S
Stability (C5)			
G_ϕ	molar Gibbs energy of species phi	$\mathrm{J\,mol^{-1}}$	S
$\mu_{\mathrm{H_2O}}$	chemical potential of water	$\mathrm{J\,mol^{-1}}$	S
n_O	number of O atoms per formula unit	–	S
n_H	number of H atoms per formula unit	–	S
n_M	number of non-O, non-H atoms	–	S
Q	species charge in units of e	–	S
Ψ_ϕ	Pourbaix potential per metal atom	$\mathrm{J\,mol^{-1}}$	S
Ψ_hull	lower envelope of Psi (the hull)	$\mathrm{J\,mol^{-1}}$	S
ΔG_pbx	energy above the aqueous Pourbaix hull	$\mathrm{J\,mol^{-1}}$	S
n_Li	number of Li atoms per formula unit	–	C
μ°_Li	Li chemical potential in Li metal	$\mathrm{J\,mol^{-1}}$	C
μ_Li	Li chemical potential at the electrode potential	$\mathrm{J\,mol^{-1}}$	C
Φ_ϕ	Li-open grand potential	$\mathrm{J\,mol^{-1}}$	C
ΔG_r	reaction free energy against Li metal	$\mathrm{J\,mol^{-1}}$	C
Δn	Li released by the reaction	–	C
$\Delta\Phi$	grand-potential change of the reaction	$\mathrm{J\,mol^{-1}}$	C
E^*	potential limit set by one reaction	V	C
E_red	lower limit of the stability window	V	C
E_ox	upper limit of the stability window	V	C
Insulating film (C4, R-cell)			
d	film thickness	m	C
ρ_e	film electronic resistivity	$\Omega\,\mathrm{m}$	C
R_A	area-specific film resistance	$\Omega\,\mathrm{m^2}$	C
$R_{A,\mathrm{max}}$	admissible area-specific resistance (input)	$\Omega\,\mathrm{m^2}$	C
ϕ_B	tunnelling barrier height	J	C
κ	tunnelling decay constant	$\mathrm{m^{-1}}$	C
j_0	tunnelling prefactor current density	$\mathrm{A\,m^{-2}}$	C
j_req	required current density (input)	$\mathrm{A\,m^{-2}}$	C
$d^\mathrm{tun}_\mathrm{max}$	thickness bound, direct tunnelling	m	C
$d^\mathrm{ohm}_\mathrm{max}$	thickness bound, ohmic limit	m	C

Conventions. Angles in rad; free energies per formula unit (J) or per mole ($\mathrm{J\,mol^{-1}}$) as stated; activities on the molal scale; potentials vs SHE in R-slurry and vs $\mathrm{Li^+/Li}$ in R-cell. Ω is the only supersaturation variable (the charter's S for C3 is Ω). The functions W4 implements, with exact limits, monotonicity and the sign-flip tests, are in `theory/contract.yaml`.